Class test (20 marks)

Introduction to Statistics

Exercise 1

The sample median of the initial 99 values of a data set consisting of 198 values is equal to 120, whereas the sample median of the final 99 values is equal to 100. What can you conclude about the sample median of the entire data set? Argue clearly.

(8 marks)

Exercise 2

Consider the Poisson random variable $X \sim \text{Poisson}(100)$. You can obtain the p.m.f. using the infinite series expansion of $e^{\lambda}, \lambda > 0$. Approximate the formula for $P(X > 20)$. Hint: Poisson has an analogous additive property like the chi-square.

(5 marks)

Exercise 3

A gambler gambles k times in a day. He wins in p proportion of the games played. Both k, p are unknown. But we have data on how many games he won everyday last month (say $n = 30$ days). We want to use this data to derive an estimator of both k and p. Use the method of moments approach to derive the forms of the estimators.

(7 marks)